

- ① No. of MAC operations
= Output dimension $\times$ Input dimension
(or) = No. of Neurons $\times$ No. of outputs (Neurons)
in current layer from previous layer.

layer 1 :	256×784	$= 200704$
layer 2 :	128×256	$= 32768$
layer 3 :	10×128	$= 1280$

- ② Total MACs for one forward pass

$$\begin{aligned}\text{Total MACs} &= 200704 + 32768 + 1280 \\ &= 234,752\end{aligned}$$

- ③ Total No. of trainable parameters
= Same as MAC calculation from ①
 $= 234,752$

- ④ Total weight memory (FP32)
= Total No. of trainable parameters $\times 4$
 $= 234,752 \times 4$
 $= 939,008 \text{ B} = 939 \text{ KB}$

5) Activation Memory

= 4 × Total No. of Neurons incl. input layer.

$$= (784 + 256 + 128 + 10) \times 4$$

$$= 1178 \times 4 = 4712 \text{ B}$$

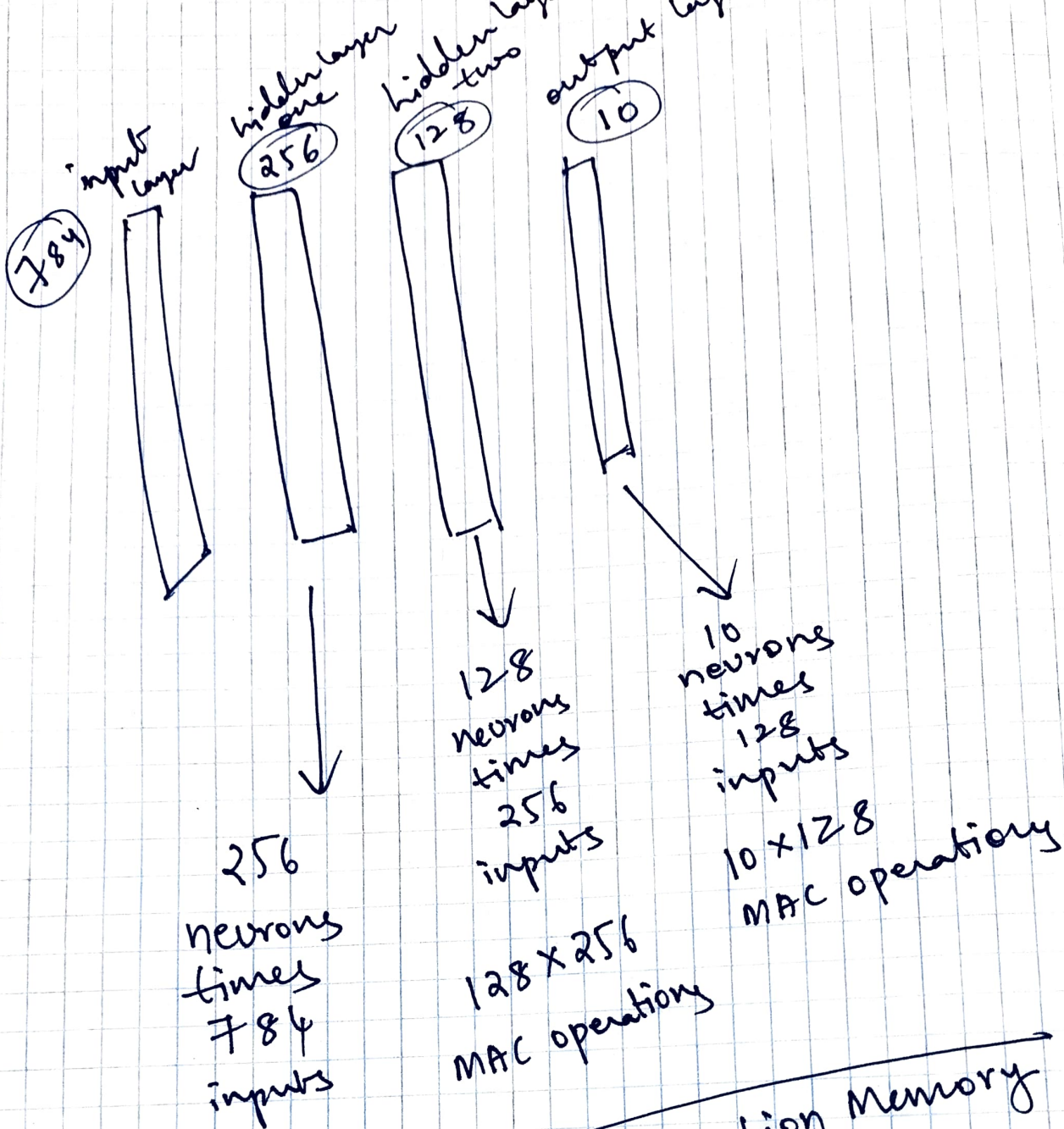
6) Arithmetic Intensity

$2 \times \text{Total MACS}$

= $\frac{\text{Weight Bytes} + \text{Activation Bytes}}$

$$= \frac{2 \times 234752}{4712 + 939008}$$

$$= 0.497503 \text{ FLOP/Byte.}$$



$$256 \times 784$$

MAC operations

Activation Memory

$$= 4 \text{ bytes} \times \text{No. of neurons}$$

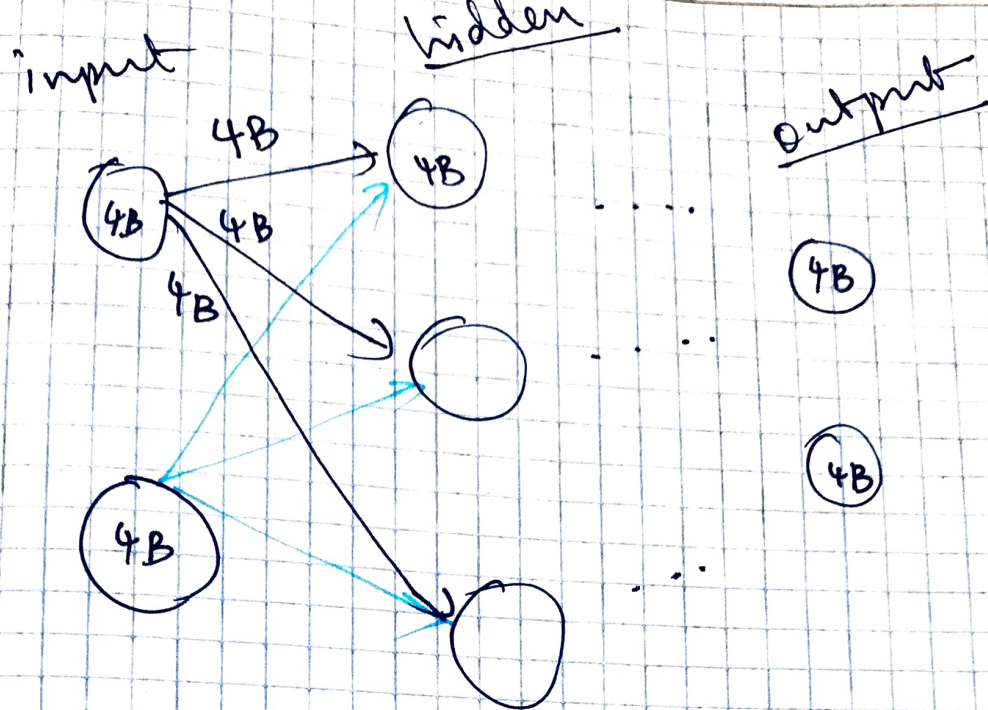
$$= 4 \times (784 + 256 + 128 + 10)$$

Weight Memory

$$= 4 \text{ bytes} \times \text{No. of connections}$$

$$= 4 \text{ bytes} \times 234752$$

$$= 939008$$



$$\text{Activation Memory} = 4712 \text{ B}$$

$$\begin{aligned} \text{Weight Memory} &= 200704 + 32768 + 128 \\ &= 234752 \times 4 \\ &= 939008 \text{ B.} \end{aligned}$$

$$\text{Total MACs} = 234,752$$

for one pass.

Arth Intensity

$$\begin{aligned} & \frac{2 \times 234752}{4712 + 939008} \\ &= 0.497503. \end{aligned}$$